

BEC305 Python Programming — Assignment 1 — Expressions, flow control and functions

Module 1: Python Basics, Flow Control and Functions

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	<<TBD>>
Due date	<<TBD>>
Items	10 problems + 3 theory questions
CO / Bloom coverage	CO1: 13 · L1–L2 0, L3–L4 10, L5–L6 3

Objective

Build reliable habits at the point where most beginner bugs are born: types at the input boundary, indentation as syntax, condition ordering, loop termination, and the local-versus-global distinction.

Learning outcomes

- Student will be able to predict the type and value of an expression before running it.
- Student will be able to choose between if/elif, while and for on the basis of what the problem requires.
- Student will be able to write functions that return values rather than print them, and explain the scope of every name used.
- Student will be able to handle bad user input with a specific exception rather than a crash or a bare except.

Part A — Problems (17 marks)

P1. Predict, in writing, the type and value of ten mixed expressions of your own construction covering `/`, `//`, `%`, string concatenation, string replication and comparison across types. Then run them and report every case where your prediction was wrong, with the reason. [C01 | L3 | 2 marks]

it is removed. [C01 | L3 | 2 marks]

P6. Using range(), print a multiplication table, a countdown that reaches zero, and every third value between 0 and 30. Explain in each case how you accounted for the excluded stop value. [C01 | L3 | 2 marks]

P7. Write three functions: one returning a value, one printing and returning None, and one with a parameter given a default. Demonstrate that only the returning one can be used inside a larger expression. [C01 | L3 | 2 marks]

P8. Construct a program in which a function reads a global, a second function shadows it with a local assignment, and a third modifies it using the global statement. Print the global after each call and explain each result. [C01 | L4 | 1 marks]

P9. Write the Guess the Number game with a secret between 1 and 20, a limited number of attempts, hotter-colder feedback, and a try/except around the input conversion. Test it with a non-numeric guess. [C01 | L4 | 1 marks]

P10. Take a program with a large try block wrapping the whole body, shrink each try to the smallest failing statement, and report what diagnostic information the change recovers. [C01 | L5 | 1 marks]

Part B — Theory (3 marks)

T1. Explain why Python makes indentation part of the syntax, and describe one class of bug this eliminates and one it introduces. [C01 | L4 | 1 marks]

T2. Argue why short-circuit evaluation of and and or must be specified behaviour rather than left to the implementation, using a guard expression as evidence. [C01 | L5 | 1 marks]

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Control-flow choice and justification	5	while, for and if/elif each chosen with an explicit reason; unreachable-clause and loop-termination demonstrations completed and analysed.	Correct choices with brief justification.	Constructs used correctly but without justification.	Separate if statements used where elif was needed, or a loop with no terminating statement.
Functions, scope and error handling	5	Functions return rather than print, scope of every name explained, exceptions caught specifically with minimal try blocks.	Mostly correct with minor scope or granularity lapses.	Functions work but print instead of returning, or try blocks are too large.	Bare except used, or global state modified without a global statement being understood.
Code readability and commentary	3	Consistent four-space indentation, descriptive names, a comment for every non-obvious line, no tabs anywhere.	Readable and consistent with occasional missing comments.	Runs but names are terse and comments sparse.	Mixed tabs and spaces, single-letter names throughout, no comments.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO1	20
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.